

Operating System Security Fundamentals – Kali Linux (Detailed Explanation)

This document provides a detailed explanation of Operating System Security fundamentals using Kali Linux. It explains what security measures were applied, why they are necessary, and how they help reduce the operating system attack surface.

1. User Accounts and Privilege Management

Linux uses a privilege-based security model. Standard users are intended for daily activities, while administrative privileges are reserved for system-level tasks. Using a standard user with sudo access reduces the risk associated with running all processes as root.

2. File Permissions and Ownership

File permissions define who can read, write, or execute files. Improper permissions can expose sensitive data or allow unauthorized modification. Commands such as `ls -l`, `chmod`, and `chown` help enforce least privilege.

3. Processes and Services

Processes are running instances of programs, while services are background processes that start automatically. Each active service increases the system attack surface and must be reviewed for necessity.

4. Disabling Unnecessary Services

Disabling services that are not required reduces the number of potential entry points attackers can exploit, thereby strengthening overall system security.

5. Firewall Configuration

Firewalls control inbound and outbound network traffic. Enabling UFW and applying secure default rules helps prevent unauthorized access to the system.

6. Log Monitoring

Kali Linux uses `systemd journal` for logging instead of traditional `auth.log` files. Reviewing logs with `journalctl` provides visibility into authentication events and system activity.

7. Operating System Hardening Best Practices

OS hardening involves applying security configurations such as updates, least privilege, firewall usage, service minimization, and continuous monitoring to reduce system exposure.

8. Conclusion

Operating system security is a fundamental requirement in cybersecurity. Applying proper access control, permission management, service hardening, firewall rules, and log monitoring significantly improves system resilience against attacks.